1 Sum and difference of cubes

The sum or difference of two cubes can be factored into the product of a binomial (two terms) times a trinomial (three terms).

Difference of cubes:

$$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$$

Sum of cubes:

$$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$$

Examples:

1. Factorise $27x^3 - y^3$

   Write as a difference of two cubes:

   $(3x)^3 - y^3$

   Then factorise:

   $(3x - y)\Big((3x)^2 + 3xy + y^2\Big) = (3x - y)(9x^2 + 3xy + y^2)$

2. Factorise $40a^3 + 625b^3$

   Factorise out a factor of 5 first:

   $5(8a^3 + 125b^3)$

   Write as a sum of cubes:

   $5\Big((2a)^3 + (5b)^3\Big)$

   Factorise:

   $5(2a + 5b)\Big((2a)^2 - 2a \times 5b + (5b)^2\Big) = 5(2a + 5b)(4a^2 - 10ab + 25b^2)$

Questions

(Answers - page **??**)

1. Factorise the following:

   (a) $x^3 - 64$

   (b) $x^3 + 27$

   (c) $3x^3 - 24$

   (d) $27x^3 - 8$

   (e) $8x^3 + 125$

2. Let ω be a non-real cube root of unity, such that $\omega^3 = 1$ and $\omega \neq 1$.
   Prove that $1 + \omega + \omega^2 = 0$

3. Prove the trigonometric identity:
   $\frac{\sin^3(x)+\cos^3(x)}{\sin(x)+\cos(x)} = 1 - \frac{1}{2}\sin(2x)$

4. Find the indefinite integral:
   $\int \frac{x^3+8}{x+2}\,dx$

5. Find all real solutions to the equation:
   $x^6 - 7x^3 - 8 = 0$